

Money Growth and Inflation

Mbak 9515: Economics

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Readings

- Chapter 12. Money Growth and Inflation
 - ▶ The classical theory of inflation
 - ▶ Classical dichotomy
 - ▶ Quantity theory of money
 - ▶ The costs of inflation

Review

- **Inflation**

- ▶ Increase in the overall level of prices

- **Deflation**

- ▶ Decrease in the overall level of prices

- **Hyperinflation**

- ▶ Extraordinarily high rate of inflation
- ▶ e.g.) Germany, Zimbabwe, and Venezuela

hyperinflation

Inflation



- 2005 to 2018: Prices rose at an average rate of **1.2%** per year
- The 1970s: Prices rose by **7.8%** per year

The Classical Theory of Inflation

- **Classical theory of money**

- ▶ Quantity theory of money
- ▶ Explain the long-run determinants of the price level
- ▶ Explain the inflation rate

- Why do people worry about inflation?

- ▶ **Price level:** a measure of the overall level of prices at a particular point in time as measured by a price index such as the CPI
- ▶ **Relative price:** the price of a specific good or service *in comparison to* the prices of other goods and services
- ▶ Inflation: *rise* in the price level $\Rightarrow$ *Lower value of money*

The Classical Theory of Inflation

- **Money demand**

- ▶ Reflects how much wealth people want to hold in liquid form
- ▶ Depends on
 - ★ Credit cards
 - ★ Availability of ATM machines
 - ★ Interest rate
 - ★ Average level of prices in economy

- Demand curve – downward sloping

The Classical Theory of Inflation

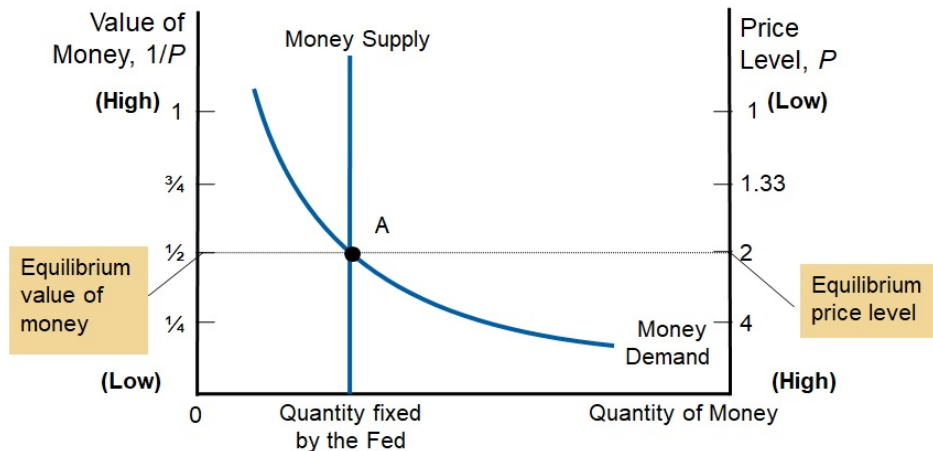
- **Money supply**

- ▶ Determined by the Fed and the banking system
- ▶ *Supply curve is vertical*

- **In the long run**

- ▶ Money supply and money demand are brought into equilibrium by the overall level of prices

The Classical Theory of Inflation



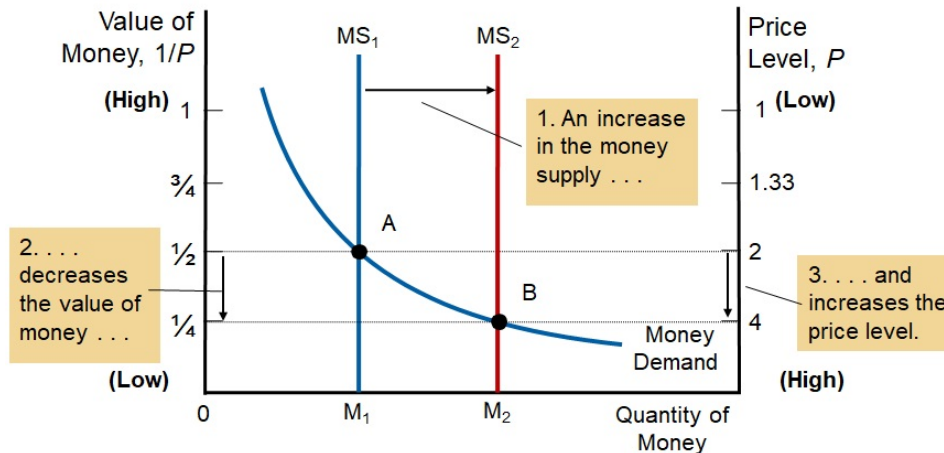
Effects of a Monetary Injection

- Economy is in equilibrium
 - ▶ If the Fed doubles the supply of money
 - ★ Prints bills
 - ★ Drops them on market
 - ▶ Or the Fed: open-market purchase
 - ▶ New equilibrium
 - ★ Supply curve shifts *right*
 - ★ Value of money decreases
 - ★ Price level increases

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Effects of a Monetary Injection

Quantity theory of money

- The quantity of money available in the economy determines (the value of money) the price level
- Growth rate in quantity of money available determines the inflation rate
- Adjustment process
 - ▶ Excess supply of money
 - ▶ Increase in demand of goods and services
 - ▶ Price of goods and services increases
 - ▶ Increase in price level
 - ▶ Increase in quantity of money demanded
 - ▶ New equilibrium

Classical Dichotomy

- Nominal variables
 - ▶ Variables measured in monetary units
- Real variables
 - ▶ Variables measured in physical units
 - ▶ Relative prices, real wages, real interest rate
- **Classical dichotomy**
 - ▶ Theoretical separation of nominal and real variables

Classical Dichotomy

- Developments in the monetary system
 - ▶ Influence nominal variables
 - ▶ Irrelevant for explaining real variables
- Monetary neutrality
 - ▶ Changes in money supply don't affect real variables
 - ▶ Not completely realistic **in short-run**
 - ▶ Correct **in the long run**

Classical Dichotomy

Short run vs Long run

- **Long run**

- ▶ the conceptual time period in which there are *no fixed* factors of production

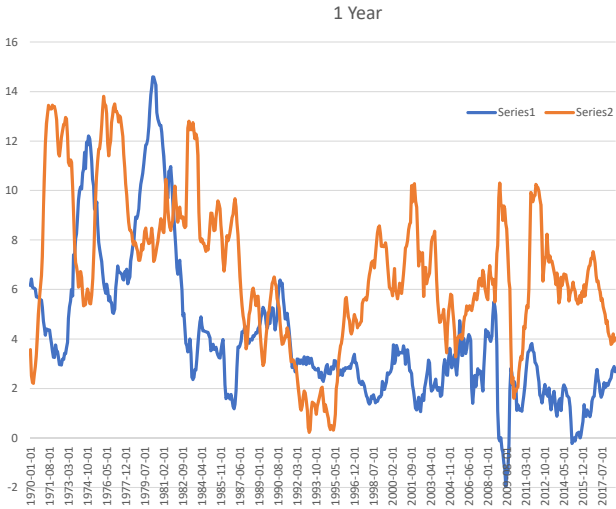
- **Short run**

- ▶ some factors are variable and others *are fixed*

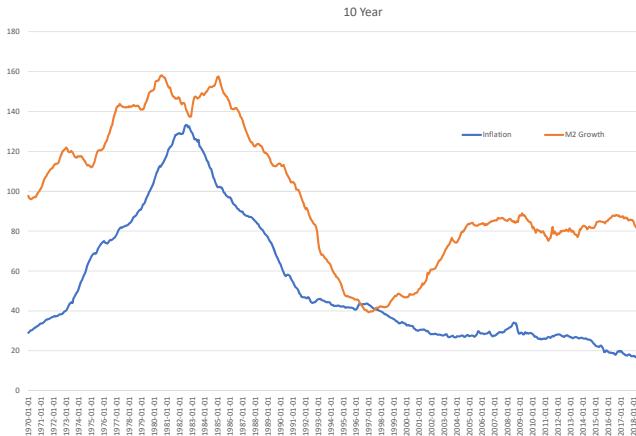
- **Why do we care about money supply?**

- ▶ **In the long run**, the amount of *money* circulating in an economy and the general level of *prices* are closely linked

Money and Inflation in the Long Run



Money and Inflation in the Long Run



Money and Inflation in the Long Run

Velocity of Money

- A measure of the speed at which money circulates
 - ▶ The speed at which money changes hands in transactions involving final goods and services
- Nominal GDP divided by the stock of money

$$\text{Velocity} = \frac{\text{Value of transactions}}{\text{Money stock}}$$

$$= \frac{\text{Nominal GDP}}{\text{Money stock}}$$

$$= \frac{P \times Y}{M}$$

- Advanced technologies and payment methods increase velocity
 - ▶ ATMs, Apple pay, Samsung pay $\Rightarrow$ people carry less cash

Money and Inflation in the Long Run

Example

- In 2016, M1 was \$3,247.9 billion, M2 was \$12,829.2 billion, and nominal GDP was \$18,624.5 billion. What is velocity of M1?
 - ▶ $V = \frac{\$18,624.5}{\$3,247.9} = 5.73$
 - ▶ What about velocity of M2?
 - ▶ $V = \frac{\$18,624.5}{\$12,829.2} = 1.45$

Money and Inflation in the Long Run

- The quantity equation

$$M \times V = P \times Y$$

- Approximately,

$$M \times \bar{V} = P \times \bar{Y}$$

- ▶ figure
- ▶ Money supply M and price level P have one-to-one relationship!
- ▶ As the Fed increases money supply 10%, price level also rises by 10%

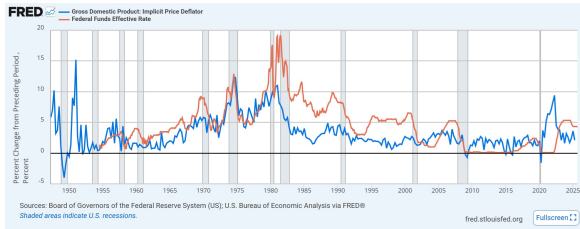
Fisher Equation

- **The nominal interest rate** is the rate of interest specified in the loan contract
- **The real interest rate** is the difference between *the nominal interest rate* and *the inflation rate*
 - ▶ It measures the reward to lenders, in terms of purchasing power, paid by borrowers for lenders giving up spending over the contract period

$$i = r + \pi$$

- ▶ where i is the nominal interest rate, r is the real interest rate, and π is the current inflation rate
- Fisher effect: the tendency for nominal interest rates to be *high* when inflation is *high* and *low* when inflation is *low*

Fisher Effect



The True Costs of Inflation

- **Noise in the price system**

- ▶ Increasing prices should *signal* increased demand
- ▶ but a high inflation environment interferes with the signal (noise)

- **Distortions of the tax system**

- ▶ Many taxes are indexed; income tax
- ▶ higher effective tax rates caused by 'bracket creep' in a progressive, non-indexed tax system

- **Shoe leather costs**

- ▶ The costs associated with holding less cash on hand and more in the bank so as to offset the reduction of purchasing power with interest

- **Unexpected redistributions of wealth**

- ▶ Inflation redistributes rather than destroys purchasing power

- **Interference with long-term planning**

- ▶ Long-run business and personal decisions become more difficult to make with high and erratic inflation

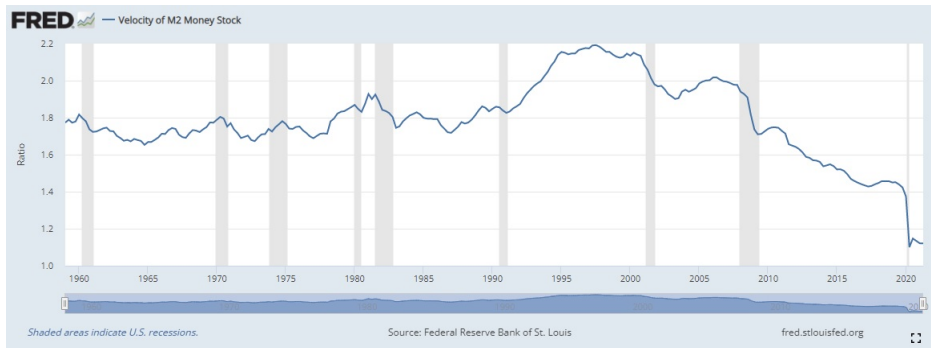
Is zero inflation a good thing?

- If high rates of money growth lead to inflation, why do countries allow their money supplies to rise quickly?
- Hyperinflation!
- Grease the wheels of the labor market
 - ▶ nominal wage increases due to inflation and this makes people supply their labor more

Hyperinflation



Velocity of Money



Goback